Paper-by-Paper Fit, Gap, and Strength Matrix

Comparison against the MoltBook project focus: thread-level sentiment homeostasis, parent-child sentiment alignment, author sentiment consistency, interaction structure, and sentiment-method robustness.

Paper Link	Fit	What Matches Our Research	Gap Our Project Can Fill	Strong Point / How To Use It
The Rise of AI Agent Communities arXiv:2602.12634	Very high	Same platform; discourse, topics, broad sentiment tone, neutral writing, positivity in community contexts, sparse interaction network, hubs, low reciprocity.	Does not deeply model parent-child sentiment alignment, thread-phase trajectories, neutral attractor behavior, negative de-escalation, author sentiment consistency, or scorer robustness.	Closest competitor and strongest anchor citation. Use it to show your study extends platform-level analysis into thread-level sentiment dynamics.
Let There Be Claws arXiv:2602.20044	High	Strong match for RQ1: directed comment graphs, reciprocity, community structure, centrality, attention inequality, hub-authority roles, memory and identity topics.	Mostly structural. It does not explain how sentiment behaves inside reply chains or whether hub agents stabilize discourse tonally.	Use as network-structure support. It strengthens your claim that your corpus patterns match larger MoltBook structural baselines.
"Humans welcome to observe" arXiv:2602.10127	High	Same platform; large-scale empirical analysis; topic taxonomy; toxicity and risk; temporal development; centralization and platform-level safeguards.	Risk and toxicity framing dominates. It does not focus on ordinary sentiment distribution, parent-child alignment, or rule-based sentiment robustness.	Use for platform context, growth, topic/risk framing, and motivation for studying safeguards in agent social networks.
Agents in the Wild arXiv:2602.13284	High	Same platform; safety and society framing; low reciprocity; shallow interaction; pro-human vs anti-human sentiment; critique of sociality claims.	Main object is safety and structural hollowness, not sentiment homeostasis or thread-level affective regulation.	Use as contrast: even if MoltBook sociality is shallow, your work shows measurable affective regularities inside reply chains.
Does Socialization Emerge in AI Agent Society? arXiv:2602.14299	High	Same platform; semantic stabilization, individual inertia, weak mutual influence, lack of shared social memory. Very relevant to author-level sentiment consistency.	Studies semantic and socialization dynamics rather than sentiment-class transitions or reply-level affective alignment.	Use to theorize author sentiment consistency as agent-level inertia or stable tonal persona rather than simple contagion.
Benchmarking Emergent Coordination in Large-Scale LLM Populations arXiv:2603.03555	Medium	MoltBook archive; role specialization, information diffusion, core-periphery structure, coordination overhead.	Coordination-focused rather than sentiment-focused. It does not test ordinary sentiment transitions or method robustness.	Use as background on large-scale agent coordination and core-periphery structure, not as a direct sentiment competitor.
Harm in AI-Driven Societies arXiv:2601.01090	High method fit	Strong methodological parallel: models posts as stimuli and comments as responses; studies whether harmful exposure predicts harmful replies.	Different platform and toxicity-only outcome. It does not study MoltBook or general positive/neutral/negative sentiment dynamics.	Use as a methodological template for parent-child or stimulus-response modeling, while emphasizing your broader sentiment classes.

Paper Link	Fit	What Matches Our Research	Gap Our Project Can Fill	Strong Point / How To Use It
Collective Behaviors and Social Dynamics in LLM Agents arXiv:2602.03775 / EACL 2026	Medium-high	AI-agent social network; homophily, social influence, toxicity, ideological leaning, polarization, longitudinal dynamics.	Chirper.ai rather than MoltBook. Broader social dynamics, not thread-level sentiment homeostasis.	Use as cross-platform evidence that LLM-agent social networks can be studied with social network and behavioral methods.
Characterizing LLM-driven Social Network: The Chirper.ai Case arXiv:2504.10286	Medium	AI-agent social network compared with human social network; posting behavior, abusive content, and social structure.	Different platform. It does not address MoltBook-specific comment threads, neutral attractor behavior, or scorer sensitivity.	Use as comparative background for how agent-only social networks differ from human social networks.
Unveiling Collective Behaviors of LLM-based Autonomous Agents Data and Information Management, 2026	Medium	Chirper-based social network analysis; centralized interaction, power-law degree patterns, homophily, machine behavior framing.	Does not address MoltBook, sentiment alignment, or scorer robustness.	Use for theoretical support on AI-agent networks, centralization, and homophily as machine-behavior phenomena.
Generative Agents arXiv:2304.03442	Background	Foundational agent society architecture; memory, reflection, planning, emergent social behavior.	Simulation rather than observed platform data. It does not compete with your empirical MoltBook analysis.	Use in the background section to explain why agent memory, planning, and emergent social behavior are relevant.